Generative-Model-Aided Grant-Free Random Access for GNSS-Independent MIMO-AFDM Satellite IoT Systems

Bo Wang, *Student Member, IEEE,* Wenpeng Jing, *Member, IEEE,* Zhiqun Hu, *Member, IEEE,* Shirui Zuo, *Student Member, IEEE,* and Changhao Liu, *Student Member, IEEE*

***Abstract*—This paper investigates grant-free random access in low Earth orbit satellite Internet-of-Things systems without global navigation satellite system (GNSS) assistance, where devices operate without time and frequency precompensation. We propose a multi-frame affine frequency division multiplexing (AFDM) transmission scheme that accommodates large differential delays and Doppler shifts across devices and analyze its input-output relationship. Based on this model, we learn a score-based generative channel prior directly from AFDM effective-channel samples, without imposing sparse or grid-aligned channel assumptions. From the learned prior, we construct a minimum mean-squared error (MMSE) channel denoiser that also quantifies estimation uncertainty. We then incorporate the denoiser into a message-passing (MP) receiver with two modules: multi-frame linear estimation (MFLE), which suppresses inter-user interference; and generative-prior-aided joint denoising (GPJD), which jointly updates channel and activity estimates. GPJD employs a density-ratio estimator that learns the likelihood ratio required for activity detection. The two modules iteratively exchange soft information for joint activity detection and channel estimation (JADCE). Numerical simulations demonstrate that the proposed receiver outperforms the baseline receiver in activity detection and channel estimation, while requiring fewer pilot frames to reach the same channel-estimation target.**

***Index Terms*—Affine frequency division multiplexing, grant-free random access, joint activity detection and channel estimation, score-based generative models.**

I. Introduction

Sixth-generation (6G) wireless networks are expected to provide ubiquitous connectivity for a massive number of Internet-of-Things (IoT) devices [1], [2]. To serve devices in remote areas with limited terrestrial coverage, low Earth orbit (LEO) satellites complement cellular networks for applications such as smart agriculture and climate monitoring [3], [4]. These power-limited devices transmit short packets sporadically, with only a small fraction active at a given time [1], [3], [5]. In grant-based random access, handshaking before transmission incurs significant signaling overhead, while the long propagation delay of terrestrial-satellite links further increases access latency [1], [2]. Accordingly, grant-free random access (GFRA) allows active devices to transmit pilots and data directly without first obtaining a grant [1], [2].

All authors are with Beijing University of Posts and Telecommunications (e-mails: wangb@bupt.edu.cn; [Wenpeng Jing: e-mail to be added]; [Zhiqun Hu: e-mail to be added]; [Shirui Zuo: e-mail to be added]; [Changhao Liu: e-mail to be added]).

Supporting GFRA over LEO links requires handling large delays and Doppler shifts caused by long propagation distances and high satellite mobility. The 3rd Generation Partnership Project (3GPP) Release 17 framework addresses these effects through device-side precompensation assisted by the global navigation satellite system (GNSS) [6]. With such precompensation, many existing satellite GFRA schemes assume perfect compensation or small residual delays and Doppler shifts at the receiver [3]. Frequent estimation and compensation increase energy consumption and reduce battery lifetime, making these schemes less suitable for power-limited remote IoT devices [3]. This motivates reducing dependence on GNSS, consistent with the GNSS-independent operation considered in the Release 19 study on satellite access [7]. With reduced reliance on device-side precompensation, the satellite receiver must accommodate large differential delays and Doppler shifts, which can cause severe inter-symbol and inter-carrier interference in conventional orthogonal frequency division multiplexing (OFDM) systems [3], [8].

Orthogonal time frequency space (OTFS) modulation exploits the quasi-static properties of channels in the delay–Doppler domain to mitigate these effects [3], [9]. In [3], multi-frame OTFS transmission handles large differential delays and Doppler shifts at the satellite receiver, allowing devices to operate without GNSS assistance or precompensation of these offsets. Affine frequency division multiplexing (AFDM) offers a low-complexity modulation structure based on one-dimensional chirp signals [8], [10]. Its modulator can be implemented using a single inverse fast Fourier transform and two stages of pointwise chirp multiplication [11].

Since device activity and channels are unknown at the receiver, AFDM-GFRA requires joint activity detection and channel estimation (JADCE) [10]. Existing methods exploit sporadic device activity to formulate JADCE as a compressed sensing (CS) problem and incorporate structured priors to model channel sparsity [3], [10], [12]. The scheme in [10] further uses a graph neural network to adaptively learn sparsity patterns, while retaining a predefined delay–Doppler grid for the sparse channel representation. In practical propagation environments, significant power leakage caused by fractional Doppler shifts and fractional time delays can weaken the sparsity of the effective channel matrix [8]. This motivates learning effective-channel priors from data to capture channel structures beyond the assumed sparse representation [1], [8].

Score-based generative models learn channel priors from

samples by estimating the score function of the channel distribution [13], [14]. In [8], score-based learning is performed directly on the AFDM effective channel to address fractional delay–Doppler off-grid leakage in channel estimation. For massive random access, [15] proposes a two-stage algorithm that first identifies active devices through a Transformer-based detector and then uses generative diffusion models for joint channel estimation and data detection. However, both approaches rely on iterative posterior sampling, which requires repeated score function evaluations and incurs substantial computational cost [8], [15].

The score-based turbo message passing (STMP) algorithm in [1] incorporates generative channel priors for JADCE without iterative posterior sampling. Through Tweedie's formulas, first- and second-order score networks provide channel estimates and variances for single-step minimum mean-squared error (MMSE) denoising under equivalent additive white Gaussian noise (AWGN) observations. For activity detection, STMP computes activity probabilities using Gaussian extrinsic messages obtained by projecting the channel posterior onto a Gaussian distribution [1]. For a non-Gaussian channel prior, however, matching the mean and variance does not generally preserve the marginal likelihood. The relation between density ratios and class probabilities [16] offers an alternative: estimating the activity likelihood ratio directly from samples, without relying on this Gaussian approximation or explicitly evaluating the marginal likelihood.

Against this background, we investigate GNSS-independent AFDM-GFRA for LEO satellite IoT systems. Our main contributions are summarized as follows:

1) We propose a multi-frame AFDM transmission scheme that accommodates large differential delays and Doppler shifts at the satellite receiver without requiring device-side precompensation. We derive the input-output relationship and show that the effective channels across frames differ only by Doppler-induced phase rotations. Additional pilot frames therefore provide more observations without increasing the number of channel coefficients to be estimated. Compared with the multi-frame OTFS representation of Shen et al. [3], the effective channel matrix in our model contains fewer coefficients.
2) We learn a score-based generative prior for AFDM effective channels without imposing sparse or grid-aligned channel assumptions. Building on Tweedie's formula and the relation between denoiser derivatives and estimation uncertainty, we develop a channel denoiser that provides both channel estimates and average variance without requiring a separately trained variance model.
3) We develop a message-passing JADCE receiver whose generative-prior-aided joint denoising (GPJD) module combines the proposed channel denoiser with a learned density-ratio estimator. The estimator enables activity detection under the learned channel prior without explicit marginal-likelihood evaluation. GPJD iteratively exchanges extrinsic messages with the multi-frame linear estimation (MFLE) module, which jointly processes the pilot frames to suppress inter-user interference. Simulations show that the proposed receiver outperforms the STMP baseline adapted to AFDM in activity detection and channel estimation, and requires fewer pilot frames to achieve the same channel-estimation accuracy.

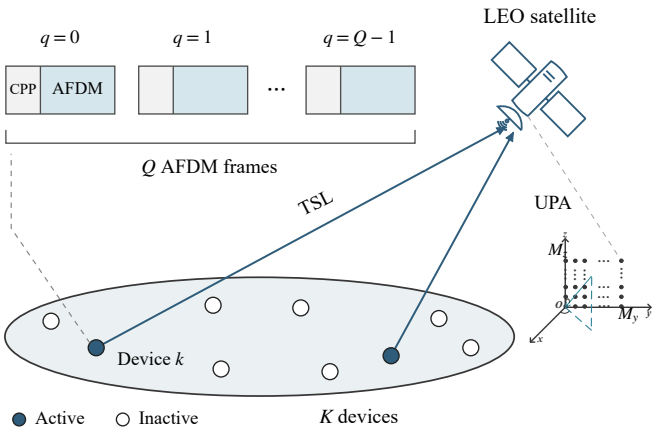


Fig. 1. LEO satellite grant-free access with multi-frame AFDM pilot transmission. The insets show the frame structure of active device k and the satellite UPA. Device activity is fixed over the Q frames. Satellite, footprint, and array artwork is adapted from [3].

The remainder of this paper is organized as follows. Section II presents the system model and the multi-frame AFDM input–output relationship. Sections III–V describe the message-passing framework, the learned channel denoiser and activity classifier, and the overall algorithm with its computational complexity analysis, respectively. Section VI presents the simulation results. Section VII concludes this paper.

Notation: Bold lowercase and uppercase letters denote vectors and matrices, respectively. $\mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes a proper complex Gaussian distribution [17]. $\|\cdot\|_F$ denotes the Frobenius norm; $(\cdot)^{\mathsf{H}}$ and $\mathrm{tr}(\cdot)$ denote conjugate transpose and trace. $\mathrm{vec}(\cdot)$ stacks matrix columns, and $\mathbf{I}_n$ is the $n \times n$ identity matrix.

II. System Model

We consider a LEO satellite-enabled grant-free random access system as illustrated in Fig. 1, where AFDM is adopted to mitigate the doubly dispersive effect [8] inherent in terrestrial-satellite links (TSLs). The system comprises K single-antenna terrestrial devices that communicate with a LEO satellite equipped with a uniform planar array (UPA) consisting of $M = M_y M_z$ antennas. During each transmission block, active devices share the same time-frequency resources to transmit their pilots to the satellite. Additionally, we assume that these terrestrial devices operate without GNSS assistance and do not pre-compensate for delay and Doppler shifts [3].

A. Terrestrial-Satellite Link Model

The TSLs experience fast variations due to the high mobility of the LEO satellite. Hence, this time variation and multipath propagation require a doubly dispersive channel model for accurate TSL characterization [3]. For stationary terrestrial devices, the Doppler shift ν_k is assumed to be common to all propagation paths within the same TSL, since the long

satellite-to-ground distance leads to only a small angular spread across multipath components [8]. The angles of arrival of these paths are approximately identical, so that they share the same array response vector $\mathbf{a}_k$ [18]. We assume that the channel in the delay-Doppler domain remains approximately constant over the Q frames of a transmission block [3]. Then, taking the start of the block as $t_{\rm c} = 0$, the channel impulse response from the kth device to the satellite after common-delay compensation by $\tau_{\rm c}$ can be represented as

$$\mathbf{h}_k(t_{\rm c}, \tau) = \mathbf{a}_k \sum_{\ell=1}^{L_k} g_{k\ell} e^{{\rm j}2\pi\nu_k t_{\rm c}} \delta(\tau - \tau_{k\ell}), \tag{1}$$

where $k = 1, \ldots, K$, and L_k is the number of physical paths. Here, $\tau_{k\ell} = \widetilde{\tau}_{k\ell} - \tau_{\rm c}$ represents the differential delay for the ℓth path from the kth device to the satellite, where $\widetilde{\tau}_{k\ell}$ and $\tau_{\rm c}$ denote the propagation delay and common delay, respectively. The common delay corresponds to the propagation time between the satellite and the closest point on Earth within the beam footprint. Additionally, $g_{k\ell}$ is the path gain of the ℓth path, $\delta(\cdot)$ is the Dirac delta function, and ${\rm j}^2 = -1$.

Then, the array response vector $\mathbf{a}_k$ can be expressed as

$$[\mathbf{a}_k]_{n_z + M_z n_y} = e^{{\rm j}2\pi(d_y n_y \mu_{y,k} + d_z n_z \mu_{z,k})/\lambda}, \tag{2}$$

where $n_z = 0, \ldots, M_z - 1$, $n_y = 0, \ldots, M_y - 1$, λ is the wavelength, and d_y and d_z are the spacings between adjacent antennas along the y- and z-axes, respectively. The directional cosines along these two axes are denoted by $\mu_{y,k}$ and $\mu_{z,k}$, respectively.

B. Transmission Scheme

The multi-frame AFDM transmission scheme is proposed to handle the large differential delay and Doppler shift. In this scheme, each active device transmits pilots over Q consecutive frames within one transmission block. While AFDM modulation and demodulation are performed separately for each frame, the receiver jointly processes the resulting observations. Specifically, in the qth AFDM frame, $q = 0, \ldots, Q-1$, the kth device transmits the known pilot vector $\mathbf{p}_{qk} \in \mathbb{C}^N$, where N is the number of chirp subcarriers shared by the devices.

We construct the pilot matrices as $\mathbf{P}_k = [\mathbf{p}_{0k}, \ldots, \mathbf{p}_{Q-1,k}]^{\mathsf{T}}$ and $\mathbf{P} = [\mathbf{P}_1, \ldots, \mathbf{P}_K] \in \mathbb{C}^{Q \times KN}$, where $(\cdot)^{\mathsf{T}}$ denotes transpose. We impose the condition that $\mathbf{P}\mathbf{P}^{\mathsf{H}} = KE_p\mathbf{I}_Q$, where $\|\mathbf{p}_{qk}\|^2 = E_p$ for all k and q [1].

Through the inverse discrete affine Fourier transform (IDAFT) [10], $\mathbf{p}_{qk}$ is mapped to the time-domain signal given by

$$s_{qk}[t] = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} [\mathbf{p}_{qk}]_n e^{{\rm j}2\pi(c_1 t^2 + c_2 n^2 + nt/N)}, \tag{3}$$

where $t = 0, \ldots, N-1$, and c_1 and c_2 are AFDM modulation parameters. The corresponding discrete affine Fourier transform (DAFT) matrix is

$$\mathbf{F}_{\rm A} = \left[\frac{1}{\sqrt{N}} e^{-{\rm j}2\pi(c_2 r^2 + rt/N + c_1 t^2)} \right]_{r,t=0}^{N-1}. \tag{4}$$

Assume that the chirp-periodic prefix (CPP) duration is greater than or equal to the maximum differential delay, i.e., $N_{\rm CPP} T_s \geq \max_{k,\ell} \tau_{k\ell}$. Here $T_s = 1/(N\Delta f)$ is the sampling interval, with Δf denoting the subcarrier spacing, and $T_{\rm frame} = (N + N_{\rm CPP})T_s$ is the duration of an AFDM frame with CPP.

Next, we use a continuous-time AFDM waveform [19] with a chirp-periodic prefix extension [11] to model fractional delays. The basis function for chirp subcarrier n at time ξT_s is given by

$$\varphi_n(\xi) = \frac{1}{\sqrt{N}} e^{{\rm j}2\pi[c_2 n^2 + c_1 \xi^2 + n w_\xi/N - \zeta_n(w_\xi) w_\xi]}, \tag{5}$$

where $w_\xi = \langle \xi \rangle_N$, $\langle x \rangle_N = x - N\lfloor x/N \rfloor$, $\zeta_n(w) = \lfloor 2c_1 w + n/N \rfloor$, and $2Nc_1 \in \mathbb{N}_+$.

After CPP removal and DAFT demodulation, the effective channel from the kth device to the satellite in frame 0 can be expressed as

$$\mathbf{H}_k = \mathbf{a}_k^{\mathsf{T}} \otimes \left[\sum_{\ell=1}^{L_k} g_{k\ell} \mathbf{G}(\tau_{k\ell}/T_s, \nu_k; N_{\rm CPP}) \right]^{\mathsf{T}}, \tag{6}$$

where $\mathbf{H}_k \in \mathbb{C}^{N \times MN}$ and $\otimes$ denotes the Kronecker product. Here, $\mathbf{G}(d, \nu; g)$ represents the response of a unit-gain path with delay dT_s and Doppler shift ν, given by

$$\mathbf{G}(d, \nu; g) = \mathbf{F}_{\rm A} \left[e^{{\rm j}2\pi\nu T_s (g+t)} \varphi_n(t-d) \right]_{t,n=0}^{N-1}. \tag{7}$$

Here $t_{\rm c} = (g+t)T_s$ for $t = 0, \ldots, N-1$ after removing the CPP, and $g = N_{\rm CPP}$ for frame 0.

In frame q, the useful samples start at time $N_{\rm CPP}T_s + qT_{\rm frame}$. Since all paths of device k share the Doppler shift ν_k, the effective channel in frame q can be expressed as

$$\mathbf{H}_k^{(q)} = e^{{\rm j}q\beta_k} \mathbf{H}_k, \tag{8}$$

where $\beta_k \equiv 2\pi\nu_k T_{\rm frame} \pmod{2\pi}$ and $\beta_k \in [-\pi, \pi)$. Under the quasi-static assumption, $\mathbf{H}_k$ remains approximately constant over the Q frames. For example, consider a circular orbit and a ground device in the orbital plane with an initial elevation angle of 50°. The simulation parameters with $Q = 16$ give a block duration of 11.73 ms and absolute Doppler and delay variations of approximately 3.33 Hz and 0.174 μs. The delay variation is below one-tenth of T_s, while retaining the initial Doppler shift gives a phase deviation of approximately 0.123 rad at the block end.

Absorbing the phase rotation $e^{{\rm j}q\beta_k}$ into the pilots gives the effective pilot matrix

$$\mathbf{A}_k = {\rm diag}\{e^{{\rm j}q\beta_k}\}_{q=0}^{Q-1} \mathbf{P}_k, \tag{9}$$

where $\mathbf{A} = [\mathbf{A}_1, \ldots, \mathbf{A}_K]$.

It is observed in (6) that the effective channel at each receive antenna is represented as an $N \times N$ matrix, corresponding to one block of $\mathbf{H}_k$. The multi-frame OTFS scheme in [3] uses an $L \times N \times N$ tensor for N subcarriers and L OFDM symbols per frame. The dense AFDM channel representation contains MN^2 channel entries per device, compared with MLN^2 in that OTFS representation. With the same subcarrier spacing and prefix length, L AFDM frames span the duration of one OTFS frame and contain the same number of useful samples.

C. Problem Formulation

Each device determines whether to access the channel in an independent and identically distributed (i.i.d.) manner. We use an indicator variable α_k to model the device activity as

$$\alpha_k = \begin{cases} 1, & \text{if the } k\text{th device is active,} \\ 0, & \text{otherwise.} \end{cases}$$

Let ρ represent the probability of a device being active, i.e., $\Pr(\alpha_k = 1) = \rho$, where $\rho \ll 1$. As a result, only a small subset of the devices is active in any given transmission block. Each device's activity state remains fixed throughout the block.

Let $\mathbf{X}_k = \alpha_k \mathbf{H}_k$. Denote by $\mathbf{Y}_q \in \mathbb{C}^{M\times N}$ the received signal during the qth AFDM frame after CPP removal and DAFT demodulation. Define $\mathbf{Y} = [\mathrm{vec}(\mathbf{Y}_0^{\mathsf{T}}), \ldots, \mathrm{vec}(\mathbf{Y}_{Q-1}^{\mathsf{T}})]^{\mathsf{T}}$ and $\mathbf{X} = [\mathbf{X}_1^{\mathsf{T}}, \ldots, \mathbf{X}_K^{\mathsf{T}}]^{\mathsf{T}}$. We express the received signal $\mathbf{Y}$ over the Q AFDM frames in a more compact form as

$$\mathbf{Y} = \sum_{k=1}^{K} \mathbf{A}_k \mathbf{X}_k + \mathbf{W} = \mathbf{A}\mathbf{X} + \mathbf{W}, \tag{10}$$

where $\mathbf{Y}, \mathbf{W} \in \mathbb{C}^{Q\times MN}$, $\mathbf{A} \in \mathbb{C}^{Q\times KN}$, and $\mathbf{X} \in \mathbb{C}^{KN\times MN}$. The elements of $\mathbf{W}$ are drawn i.i.d. from $\mathcal{CN}(0, \sigma^2)$. Due to the sporadic access of the devices, matrix $\mathbf{X}$ exhibits a block-sparsity structure. Specifically, if the kth device is inactive, we have $\mathbf{X}_k = \alpha_k \mathbf{H}_k = \mathbf{0}$ [1].

Since the Q frames share the same $\mathbf{X}$, additional pilot frames provide more observations without increasing the number of unknown channel coefficients. The linear system in (10) is underdetermined when $Q < KN$.

In this paper, we aim to simultaneously detect the active devices and estimate the corresponding channels based on the observation $\mathbf{Y}$ under a Bayesian inference framework. Fractional delays and Doppler shifts cause power leakage that can weaken the sparsity of $\mathbf{H}_k$ [8]. We therefore use the training-channel distribution $q_{\mathrm{tr}}(\mathbf{H}_k)$ as a prior on the full channel matrix, without imposing sparsity or factorizing the prior element-wise. We assume that the pairs $(\mathbf{H}_k, \alpha_k)$ are independent across devices, and that $\mathbf{H}_k$ is independent of α_k.

According to Bayes' theorem and these independence relationships, the joint posterior distribution conditioned on the estimated effective pilot matrix $\widehat{\mathbf{A}}$ and noise variance $\widehat{\sigma}^2$ can be expressed as [1]

$$\begin{aligned} &q(\mathbf{X}, \{\mathbf{H}_k\}_{k=1}^K, \{\alpha_k\}_{k=1}^K \mid \mathbf{Y}) \\ &\quad\propto \prod_{j=1}^{MN} p(\mathbf{y}_j \mid \mathbf{x}_j) \\ &\quad\times \prod_{k=1}^{K} p(\mathbf{X}_k \mid \mathbf{H}_k, \alpha_k) q_{\mathrm{tr}}(\mathbf{H}_k) p(\alpha_k), \end{aligned} \tag{11}$$

where $\mathbf{x}_j$ and $\mathbf{y}_j$ denote the jth column of $\mathbf{X}$ and $\mathbf{Y}$, respectively. Here $p(\mathbf{y}_j \mid \mathbf{x}_j) = \mathcal{CN}(\mathbf{y}_j; \widehat{\mathbf{A}}\mathbf{x}_j, \widehat{\sigma}^2 \mathbf{I}_Q)$, $p(\mathbf{X}_k \mid \mathbf{H}_k, \alpha_k)$ enforces $\mathbf{X}_k = \alpha_k \mathbf{H}_k$, and $p(\alpha_k) = \rho^{\alpha_k}(1-\rho)^{1-\alpha_k}$. The inference targets are the MMSE estimate $\mathbb{E}_q[\mathbf{X}_k \mid \mathbf{Y}]$ and the posterior activity probability $q(\alpha_k = 1 \mid \mathbf{Y})$.

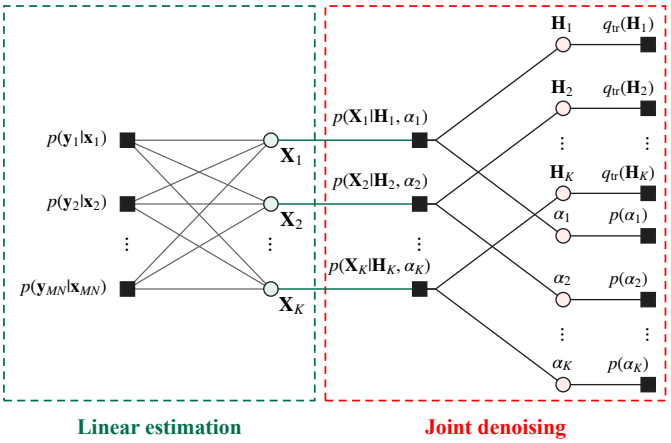


Fig. 2. Factor graph of (11) using device blocks $\mathbf{X}_k$ as variable nodes. Each likelihood factor depends on the corresponding column entries from all device blocks. Circles and squares denote variables and factors. The channel and activity prior branches are adapted from [1].

III. Message Passing Framework

In this section, we develop a message-passing (MP) framework [1] based on the combined use of the sum-product rule and the expectation propagation (EP) algorithm [20]. The framework interprets the message updates on the factor graph in Fig. 2 as iterations between two modules, namely, a linear estimator and a joint denoiser. The linear estimator separately estimates each column of $\mathbf{X}$ (i.e., $\mathbf{x}_j$) by exploiting the likelihood $p(\mathbf{y}_j \mid \mathbf{x}_j)$ and the messages from the joint denoiser. The joint denoiser estimates each device block $\mathbf{X}_k$ by exploiting the priors $q_{\mathrm{tr}}(\mathbf{H}_k)$ and $p(\alpha_k)$, along with the messages from the linear estimator.

A. Linear Estimation

Following Turbo-CS [21], we start by expressing the message to the factor node $p(\mathbf{y}_j \mid \mathbf{x}_j)$ as $\mathcal{CN}(\mathbf{x}_j; \mathbf{x}_j^{\mathrm{pri}}, \gamma \mathbf{I}_{KN})$, where $\mathbf{x}_j^{\mathrm{pri}}$ is the jth column of $\mathbf{X}^{\mathrm{pri}}$ and $\gamma > 0$. Combining this message with the likelihood $p(\mathbf{y}_j \mid \mathbf{x}_j)$ gives a Gaussian belief whose mean and covariance are given by [22]:

$$\mathbf{X}_{\mathrm{L}} = \mathbf{X}^{\mathrm{pri}} + \gamma \widehat{\mathbf{A}}^{\mathsf{H}} (\widehat{\sigma}^2 \mathbf{I}_Q + \gamma \widehat{\mathbf{A}}\widehat{\mathbf{A}}^{\mathsf{H}})^{-1} (\mathbf{Y} - \widehat{\mathbf{A}}\mathbf{X}^{\mathrm{pri}}), \tag{12}$$

$$\mathbf{\Sigma}_{\mathrm{L}} = (\gamma^{-1}\mathbf{I}_{KN} + \widehat{\sigma}^{-2}\widehat{\mathbf{A}}^{\mathsf{H}}\widehat{\mathbf{A}})^{-1}. \tag{13}$$

Eqns. (12) and (13) give, respectively, the posterior mean matrix and per-column covariance from the linear MMSE (LMMSE) estimator of $\mathbf{X}$. We further approximate the covariance matrix using a shared scalar variance $t_{\mathrm{L}} = \mathrm{tr}(\mathbf{\Sigma}_{\mathrm{L}})/(KN)$, resulting in a belief of the form $\mathcal{CN}(\mathbf{x}_j; \mathbf{x}_{\mathrm{L},j}, t_{\mathrm{L}}\mathbf{I}_{KN})$, where $\mathbf{x}_{\mathrm{L},j}$ is the jth column of $\mathbf{X}_{\mathrm{L}}$.

The extrinsic message is proportional to the ratio of the above Gaussian belief to the prior message. The variance v_{L} and mean matrix $\mathbf{R}_{\mathrm{L}}$ are respectively given by

$$v_{\mathrm{L}} = (t_{\mathrm{L}}^{-1} - \gamma^{-1})^{-1}, \tag{14}$$

$$\mathbf{R}_{\mathrm{L}} = v_{\mathrm{L}} \left(\frac{\mathbf{X}_{\mathrm{L}}}{t_{\mathrm{L}}} - \frac{\mathbf{X}^{\mathrm{pri}}}{\gamma} \right). \tag{15}$$

The extrinsic message flows rightward through the factor graph unchanged, serving as the input to the joint denoiser, with $\mathbf{R}_k$ denoting the kth device block of $\mathbf{R}_{\mathrm{L}}$ and $v = v_{\mathrm{L}}$.

B. Joint Channel Estimation and Activity Detection

For device k, the Gaussian message from the linear estimator, with mean $\mathbf{R}_k$ and variance v, gives the local observation model

$$\begin{aligned} \mathbf{R}_k &= \alpha_k \mathbf{H}_k + \mathbf{W}_k, \\ \mathrm{vec}(\mathbf{W}_k) &\sim \mathcal{CN}(\mathbf{0}, v\mathbf{I}_D), \end{aligned} \tag{16}$$

where $\mathbf{H}_k \sim q_{\mathrm{tr}}$ and $D = MN^2$.

1) Channel Estimation: Under the active hypothesis $\alpha_k = 1$, the Gaussian message from the linear estimator has the form of the likelihood $p(\mathbf{R}_k \mid \mathbf{H}_k, \alpha_k = 1)$ in the AWGN model (16). Computing the conditional mean of $\mathbf{H}_k$ therefore amounts to MMSE denoising. The channel mean and a shared average variance are approximated by

$$\mathbf{F}_k \approx \mathbb{E}[\mathbf{H}_k \mid \mathbf{R}_k, \alpha_k = 1], \tag{17}$$

$$\tau^{\mathrm{post}} \approx \frac{1}{KD} \sum_{k=1}^{K} \mathrm{tr}\,\mathrm{Cov}(\mathrm{vec}(\mathbf{H}_k) \mid \mathbf{R}_k, \alpha_k = 1). \tag{18}$$

Calculating the channel mean and shared average variance involves high-dimensional integrals.

We approximate the channel prior by a Gaussian site $\mathcal{CN}(\mathrm{vec}(\mathbf{H}_k); \mathrm{vec}(\mathbf{E}_k), \tau_k^{\mathrm{ext}}\mathbf{I}_D)$ [23]. Dividing the Gaussian belief, with mean $\mathbf{F}_k$ and variance τ^{post}, by the input likelihood gives the site parameters

$$\tau_k^{\mathrm{ext}} = ((\tau^{\mathrm{post}})^{-1} - v^{-1})^{-1}, \tag{19}$$

$$\mathbf{E}_k = \tau_k^{\mathrm{ext}} \left(\frac{\mathbf{F}_k}{\tau^{\mathrm{post}}} - \frac{\mathbf{R}_k}{v} \right). \tag{20}$$

2) Activity Detection: Marginalizing over $\mathbf{H}_k$ in (16) yields the likelihoods conditioned on $\alpha_k = 1$ and $\alpha_k = 0$, respectively:

$$q_{1,v}(\mathbf{R}) = \mathbb{E}[\mathcal{CN}(\mathrm{vec}(\mathbf{R}); \mathrm{vec}(\mathbf{H}), v\mathbf{I}_D)], \tag{21}$$

$$q_{0,v}(\mathbf{R}) = \mathcal{CN}(\mathrm{vec}(\mathbf{R}); \mathbf{0}, v\mathbf{I}_D). \tag{22}$$

The expectation is taken over $\mathbf{H} \sim q_{\mathrm{tr}}$.

The belief of α_k given $\mathbf{R}_k$ is a Bernoulli distribution with parameter $p^\star(\mathbf{R}_k, v)$, where

$$p^\star(\mathbf{R}, v) = \frac{\rho q_{1,v}(\mathbf{R})}{(1-\rho) q_{0,v}(\mathbf{R}) + \rho q_{1,v}(\mathbf{R})}. \tag{23}$$

We denote the approximation to $p^\star(\mathbf{R}_k, v)$ used in joint estimation by p_k.

3) Joint Estimation and Extrinsic Message: Combining the site with the current Gaussian likelihood gives the conditional channel variance and mean:

$$t_k = \frac{\tau_k^{\mathrm{ext}} v}{\tau_k^{\mathrm{ext}} + v}, \tag{24}$$

$$\mathbf{H}_k^{\mathrm{post}} = \frac{\tau_k^{\mathrm{ext}} \mathbf{R}_k + v\mathbf{E}_k}{\tau_k^{\mathrm{ext}} + v}. \tag{25}$$

The approximate belief of $\mathbf{X}_k = \alpha_k \mathbf{H}_k$ is a Bernoulli–Gaussian pdf [24], with activity probability p_k and a Gaussian active-channel component having mean $\mathbf{H}_k^{\mathrm{post}}$ and covariance $t_k \mathbf{I}_D$. The Gaussian projection gives the mean of $\mathbf{X}_k$ with respect to the belief as

$$\widehat{\mathbf{X}}_k = p_k \mathbf{H}_k^{\mathrm{post}}. \tag{26}$$

The variances of the elements of $\mathbf{X}_k$ with respect to the belief are averaged over all devices and channel coefficients to obtain

$$\bar{t}_X = \frac{1}{K} \sum_k \left[p_k t_k + p_k(1-p_k) \frac{\left\| \mathbf{H}_k^{\mathrm{post}} \right\|_F^2}{D} \right]. \tag{27}$$

Based on the above, the projected Gaussian belief is expressed as $\mathcal{CN}(\mathrm{vec}(\mathbf{X}_k); \mathrm{vec}(\widehat{\mathbf{X}}_k), \bar{t}_X \mathbf{I}_D)$.

Finally, we calculate the message from the factor node $p(\mathbf{X}_k \mid \mathbf{H}_k, \alpha_k)$ to the variable node $\mathbf{X}_k$ as the ratio of the projected Gaussian belief to the Gaussian message from the linear estimator. With $b = \bar{t}_X / v$, the extrinsic variance and mean are respectively given by

$$\gamma_{\mathrm{out}} = \frac{vb}{1-b}, \tag{28}$$

$$\mathbf{X}_{k,\mathrm{out}} = \frac{\widehat{\mathbf{X}}_k - b\mathbf{R}_k}{1-b}. \tag{29}$$

The extrinsic message in (28)–(29) flows leftward through the factor graph unchanged, serving as the prior message of the linear estimator in the next iteration, i.e., $\mathbf{X}^{\mathrm{pri}} = [\mathbf{X}_{1,\mathrm{out}}^{\mathsf{T}}, \ldots, \mathbf{X}_{K,\mathrm{out}}^{\mathsf{T}}]^{\mathsf{T}}$ and $\gamma = \gamma_{\mathrm{out}}$.

IV. Learning Channel Denoising and Activity Detection

In this section, we introduce score-based representations of the channel mean and variance to circumvent the high-dimensional integrals in channel estimation. We then apply denoising score matching for channel denoising and binary classification for activity detection.

A. Channel Mean Estimation with a Score-Based Generative Prior

Under the active hypothesis in (16), $\mathbf{H} \sim q_{\mathrm{tr}}$ is observed in independent proper complex Gaussian noise with covariance $v\mathbf{I}_D$, where $v > 0$. Stacking the real and imaginary parts of $\mathrm{vec}(\mathbf{H})$ and $\mathrm{vec}(\mathbf{R})$ into $\mathbf{h}_{\mathbb{R}}$ and $\mathbf{u}$, respectively, we rewrite the AWGN observation model as

$$\mathbf{u} = \mathbf{h}_{\mathbb{R}} + \sqrt{v/2}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{2D}). \tag{30}$$

At fixed v, let $q_v^{\mathbb{R}}$ denote the density of $\mathbf{u}$. For the model in (30), Tweedie's formula [25], [26] gives the conditional mean as

$$\mathbf{m}_{\mathbb{R}}(\mathbf{u}, v) = \mathbf{u} + \frac{v}{2} \nabla_{\mathbf{u}} \log q_v^{\mathbb{R}}(\mathbf{u}), \tag{31}$$

where $\nabla_{\mathbf{u}} \log q_v^{\mathbb{R}}$ is the first-order score function of $q_v^{\mathbb{R}}$.

For a specific value of v, our goal is to learn a first-order score network $\mathbf{s}_\theta(\cdot, v) : \mathbb{R}^{2D} \to \mathbb{R}^{2D}$ parameterized by θ with the following score matching objective:

$$\min_\theta \ \mathbb{E}_{\mathbf{u} \sim q_v^{\mathbb{R}}} \left[\left\| \mathbf{s}_\theta(\mathbf{u}, v) - \nabla_{\mathbf{u}} \log q_v^{\mathbb{R}}(\mathbf{u}) \right\|^2 \right].$$

This objective cannot be directly used as the loss function since $\nabla_{\mathbf{u}} \log q_v^{\mathbb{R}}(\mathbf{u})$ is unknown. The above objective has the same minimizers as the denoising score matching objective [27], whose target is the conditional Gaussian score $-2(\mathbf{u} - \mathbf{h}_{\mathbb{R}})/v$.

In practice, the score-based MMSE denoiser is required to operate across a range of noise levels. We therefore combine the denoising score matching objectives for different values of v with weight $v/(4D)$ to establish a unified objective:

$$\mathcal{L}_{\mathrm{DSM}}(\theta) = \mathbb{E}\left[\frac{v}{4D}\left\|\mathbf{s}_\theta(\mathbf{u}, v) + \frac{2}{v}(\mathbf{u} - \mathbf{h}_{\mathbb{R}})\right\|^2\right]. \tag{32}$$

The expectation is taken over the clean channels, noise variances, and Gaussian noise.

We parameterize the score by a denoiser $\mathbf{f}_\theta(\mathbf{u}, v)$ as $\mathbf{s}_\theta = 2(\mathbf{f}_\theta - \mathbf{u})/v$. Substituting this expression into (32) yields

$$\mathcal{L}_{\mathrm{ch}}(\theta) = \mathbb{E}\left[\frac{1}{Dv}\left\|\mathbf{f}_\theta(\mathbf{u}, v) - \mathbf{h}_{\mathbb{R}}\right\|^2\right]. \tag{33}$$

The conditional mean $\mathbf{m}_{\mathbb{R}}(\mathbf{u}, v)$ minimizes this loss over all estimators of $\mathbf{h}_{\mathbb{R}}$ from $(\mathbf{u}, v)$. Reshaping the real and imaginary outputs gives the complex-valued estimate $\mathbf{F}_\theta(\mathbf{R}, v)$.

B. Channel Variance Estimation

Likewise, the conditional covariance matrix $\mathbf{P}_{\mathbb{R}} = \mathrm{Cov}(\mathbf{h}_{\mathbb{R}} \mid \mathbf{u}, v)$ can be obtained through the second-order Tweedie's formula [26] as

$$\mathbf{P}_{\mathbb{R}} = \frac{v}{2}\mathbf{I}_{2D} + \frac{v^2}{4}\nabla_{\mathbf{u}}^2 \log q_v^{\mathbb{R}}(\mathbf{u}). \tag{34}$$

Differentiating (31) with respect to $\mathbf{u}$ at fixed v gives the following compact expressions for the covariance in (34) and its normalized trace:

$$\mathbf{P}_{\mathbb{R}} = \frac{v}{2}\mathbf{J}_m, \tag{35}$$

$$\tau^\star = \frac{v}{2D}\,\mathrm{div}\,\mathbf{m}_{\mathbb{R}}. \tag{36}$$

Here $\mathbf{J}_m = \partial \mathbf{m}_{\mathbb{R}}/\partial \mathbf{u}$ denotes the Jacobian of $\mathbf{m}_{\mathbb{R}}$, and $\tau^\star$ is the average conditional variance per complex coordinate. This suggests that the average conditional variance can be estimated by computing the divergence of the trained denoiser $\mathbf{f}_\theta$ [28].

The divergence is estimated by a Monte Carlo method [29], using L_z independent random sign vectors $\mathbf{z}_\ell$ with $\mathbb{E}[\mathbf{z}_\ell \mathbf{z}_\ell^{\mathsf{T}}] = \mathbf{I}_{2D}$. Central finite differences along $\mathbf{w}_\ell = \sqrt{v/2}\,\mathbf{z}_\ell$, with step $\delta_{\mathrm{FD}} > 0$, give an estimate of the average conditional variance:

$$\widehat{\tau}_\theta(\mathbf{R}, v) = \frac{1}{2\delta_{\mathrm{FD}} D L_z}\sum_{\ell=1}^{L_z} \mathbf{w}_\ell^{\mathsf{T}} \Delta_\ell \mathbf{f}_\theta, \tag{37}$$

where $\Delta_\ell \mathbf{f}_\theta = \mathbf{f}_\theta(\mathbf{u} + \delta_{\mathrm{FD}}\mathbf{w}_\ell, v) - \mathbf{f}_\theta(\mathbf{u} - \delta_{\mathrm{FD}}\mathbf{w}_\ell, v)$.

C. Activity Probability Estimation Based on Density Ratios

The density ratio $q_{1,v}/q_{0,v}$ can be estimated from samples with binary labels $a \in \{0, 1\}$, where $q_{1,v}$ and $q_{0,v}$ are the class-conditional densities at fixed v [16]. For the activity prior ρ, Bayes' rule implies

$$\log\frac{p^\star(\mathbf{R}, v)}{1 - p^\star(\mathbf{R}, v)} = \log\frac{\rho}{1-\rho} + \log\frac{q_{1,v}(\mathbf{R})}{q_{0,v}(\mathbf{R})}. \tag{38}$$

For training, we set $\Pr(a = 1) = 1/2$. Samples from the active class $a = 1$ are generated as $\mathbf{R} = \mathbf{H} + \mathbf{W}$, with $\mathbf{H} \sim q_{\mathrm{tr}}$, whereas samples from the inactive class $a = 0$ are generated as $\mathbf{R} = \mathbf{W}$. The noise satisfies $\mathrm{vec}(\mathbf{W}) \sim \mathcal{CN}(\mathbf{0}, v\mathbf{I}_D)$ independently of $\mathbf{H}$, and both classes use the same sampling distribution of v. We train the classifier with parameters ϕ by minimizing the binary cross-entropy loss

$$\mathcal{L}_{\mathrm{act}}(\phi) = \mathbb{E}[\log(1 + e^{z_\phi}) - a z_\phi], \tag{39}$$

where $z_\phi = z_\phi(\mathbf{R}, v, \mathbf{F}_\theta(\mathbf{R}, v))$. For the balanced training distribution, the Bayes-optimal logit is

$$z^\star(\mathbf{R}, v) = \log\frac{q_{1,v}(\mathbf{R})}{q_{0,v}(\mathbf{R})}. \tag{40}$$

Given a trained classifier z_ϕ, we use $\ell_k = z_\phi(\mathbf{R}_k, v, \mathbf{F}_k)$ as an estimate of the log likelihood ratio, where $\mathbf{F}_k = \mathbf{F}_\theta(\mathbf{R}_k, v)$. Substituting this estimate into (38) gives the activity update

$$p_k = \mathrm{sigmoid}(\mathrm{logit}\,\rho + \ell_k), \tag{41}$$

where $\mathrm{logit}\,a = \log[a/(1-a)]$ and $\mathrm{sigmoid}\,b = (1 + e^{-b})^{-1}$.

V. GP-MP Algorithm

A. Overall Algorithm

The trained denoiser provides the channel mean $\mathbf{F}_k = \mathbf{F}_\theta(\mathbf{R}_k, v)$, while the variance estimate $\widehat{\tau}_k = \widehat{\tau}_\theta(\mathbf{R}_k, v)$ is given by (37). Averaging $\widehat{\tau}_k$ over devices and applying the positive lower bound $\tau_{\min}$ gives τ^{post}. The trained activity classifier provides p_k through (41).

Fig. 3 illustrates the generative-prior message-passing (GP-MP) receiver, in which GPJD uses the trained denoiser and activity classifier, while MFLE is derived analytically.

The phases are either known or estimated once before MP and remain fixed during the iterations. For initialization, $\widehat{P}_H$ is the normalized training-channel power, computed by averaging $\|\mathbf{H}\|_F^2/D$ over the training channels.

Algorithm 1 presents the pseudocode for GP-MP.

B. Parameter Estimation and Numerical Implementation

For noise estimation, the Wax–Kailath minimum description length (MDL) criterion [30], [31] is used to estimate the signal covariance rank $\widehat{r}_{\mathrm{s}}$ based on the eigenvalues $\lambda_1 \geq \cdots \geq \lambda_M$ of the sample covariance matrix. Let $\mathbf{y}_{qr}$ denote column r of $\mathbf{Y}_q$ and $N_{\mathrm{s}} = QN > M \geq 2$. The sample covariance and noise estimate are

$$\widehat{\mathbf{R}}_y = \frac{1}{N_{\mathrm{s}}}\sum_{q,r}\mathbf{y}_{qr}\mathbf{y}_{qr}^{\mathsf{H}}, \tag{42}$$

$$\widehat{\sigma}^2 = \frac{\sum_{i=\widehat{r}_{\mathrm{s}}+1}^{M}\lambda_i}{M - \widehat{r}_{\mathrm{s}}}. \tag{43}$$

This estimate assumes a low-rank signal covariance and is held fixed during MP.

For phase estimation, the observations are stacked into the matrix $\mathbf{Y}_{\mathrm{s}} = [\mathbf{Y}_0, \ldots, \mathbf{Y}_{Q-1}]^{\mathsf{T}} \in \mathbb{C}^{QN \times M}$. The dominant line-of-sight component of the default NTN-TDL-D channel

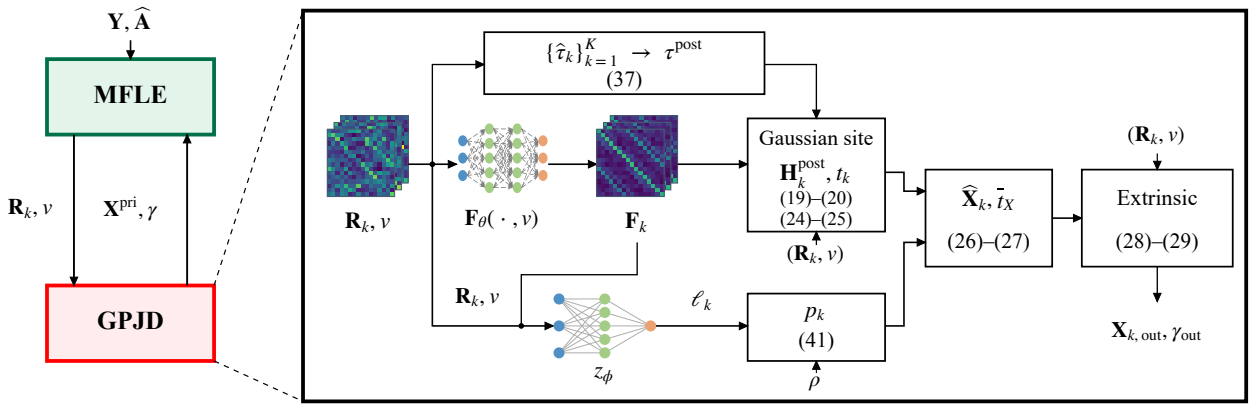


Fig. 3. Message exchange between MFLE and GPJD, with an expanded view of GPJD. The denoising illustration follows [1].

Algorithm 1 GP-MP for JADCE

Input: $\mathbf{Y}$, $\{\mathbf{p}_{qk}\}$, available phases, ρ, trained $\mathbf{f}_\theta$ and z_ϕ.
Output: $\{s\widehat{\mathbf{X}}_k, s\mathbf{H}_k^{\text{post}}, p_k, \widehat{\alpha}_k\}_{k=1}^K$.

Estimate $\widehat{\sigma}^2$; estimate any unknown phases once as described in Section V-B.
Construct $\widehat{\mathbf{A}}$ using fixed phases; normalize $\mathbf{Y}$ and $\widehat{\sigma}^2$ as in Section V-B.
Compute the SVD of $\widehat{\mathbf{A}}$; draw one random sign vector per device and reuse it across iterations.
Initialize $\mathbf{X}^{\text{pri},(0)} \leftarrow \mathbf{0}$, $\gamma^{(0)} \leftarrow \rho\widehat{P}_H$.
Initialize $\mathbf{E}_k \leftarrow \mathbf{0}$, $\tau_k^{\text{ext}} \leftarrow 1$, $\kappa^{(1)} \leftarrow 1$.
for $i = 1, \ldots, I_{\max}$ **do**
  % MFLE
  Update $\mathbf{R}_{\text{L},k}$ and v_{L} by (12)–(15), using (52)–(54).
  $\mathbf{R}_k \leftarrow \mathbf{R}_{\text{L},k}$, $\bar{v} \leftarrow v_{\text{L}}$; apply (47)–(48) for $i \geq 2$.
  $v \leftarrow \kappa^{(i)}\bar{v}$.
  % GPJD
  $\mathbf{F}_k \leftarrow \mathbf{F}_\theta(\mathbf{R}_k, v)$.
  Compute $\widehat{\tau}_k$ and p_k by (37) and (41).
  Average $\widehat{\tau}_k$ over k to obtain τ^{post}.
  Apply the lower bound $\tau_{\min}$ to τ^{post}.
  if $\tau^{\text{post}} < v$ **then**
    Update $\mathbf{E}_k$ and τ_k^{ext} by (19)–(20).
  end if
  Update $\mathbf{H}_k^{\text{post}}$ and t_k by (24)–(25).
  Update $\widehat{\mathbf{X}}_k$ and $\bar{t}_X$ by (26) and (27).
  Update $\kappa^{(i+1)}$ as in Section V-B.
  Compute $\mathbf{X}_{k,\text{out}}$ and γ_{out} by (28)–(29) with clipping.
  $\mathbf{X}_k^{\text{pri},(i)} \leftarrow \mathbf{X}_{k,\text{out}}$, $\gamma^{(i)} \leftarrow \gamma_{\text{out}}$; apply (50)–(51) for $i \geq 2$.
  Update the stopping history.
  if the stopping conditions in Section VI-A4 hold **then**
    break
  end if
end for
$\widehat{\alpha}_k \leftarrow \mathbf{1}\{p_k \geq 1/2\}$.
return $\{s\widehat{\mathbf{X}}_k, s\mathbf{H}_k^{\text{post}}, p_k, \widehat{\alpha}_k\}_{k=1}^K$.

motivates a single-path approximation of each device's channel for estimating its common Doppler shift. Absorbing the common scalar phase into the path gain gives

$$[\mathbf{h}_k^{\text{p}}(d,\nu)]_{qN+r} = e^{\text{j}2\pi\nu q T_{\text{frame}}}[\mathbf{G}_0(d,\nu)\mathbf{p}_{qk}]_r, \tag{44}$$

where $\mathbf{G}_0(d,\nu) = e^{-\text{j}2\pi c_1 d^2}\mathbf{G}(d,\nu;0)$.

With $\mathbf{h}_k^{\text{p}} = \mathbf{h}_k^{\text{p}}(d_k,\nu_k)$, the observations can be expressed as $\mathbf{Y}_{\text{s}} \approx \sum_k \mathbf{h}_k^{\text{p}}\mathbf{b}_k^{\text{T}} + \mathbf{W}_{\text{s}}$. This observation model is complemented by independent Gaussian priors over the gains, $\mathbf{b}_k \sim \mathcal{CN}(\mathbf{0}, \gamma_k\mathbf{I}_M)$, $\gamma_k \geq 0$. Phase estimation is formulated as the minimization of the negative log marginal likelihood [32], obtained by integrating out the gains:

$$\mathbf{S} = \widehat{\sigma}^2\mathbf{I}_{QN} + \sum_k \gamma_k \mathbf{h}_k^{\text{p}}\mathbf{h}_k^{\text{p},\text{H}}, \tag{45}$$

$$\mathcal{L} = \log\det\mathbf{S} + \frac{1}{M}\,\text{tr}(\mathbf{Y}_{\text{s}}^{\text{H}}\mathbf{S}^{-1}\mathbf{Y}_{\text{s}}). \tag{46}$$

The objective $\mathcal{L}$ is minimized by coordinate updates, with each γ_k solved for analytically and (d_k,ν_k) refined using L-BFGS-B [33].

The estimated Doppler determines $\widehat{\beta}_k$ and $\widehat{\mathbf{A}}$ through (8)–(9).

To match the scale used for denoiser training, the channel and observation quantities $(\mathbf{H}_k, \mathbf{X}_k, \mathbf{Y}, \mathbf{W})$ are divided by the root mean square s of the training-channel entries, and the noise variance is divided by s^2. The same symbols denote the normalized quantities.

In each iteration, the mean and variance updates are damped using a linear combination of the current update and the previous damped message [34]. With damping factor $\eta \in (0,1]$, the input messages to GPJD for $i \geq 2$ are

$$\mathbf{R}_k^{(i)} = (1-\eta)\mathbf{R}_k^{(i-1)} + \eta\mathbf{R}_{\text{L},k}^{(i)}, \tag{47}$$

$$\bar{v}^{(i)} = (1-\eta)\bar{v}^{(i-1)} + \eta v_{\text{L}}^{(i)}. \tag{48}$$

At iteration i, GPJD uses $v^{(i)} = \kappa^{(i)}\bar{v}^{(i)}$. After joint denoising, the update of the input variance follows from minimizing the expected Gaussian negative log likelihood, with the current approximate posterior held fixed [35]. Setting the derivative with respect to the variance to zero yields

$$v_{\text{tar}} = \frac{1}{KD}\sum_{k=1}^{K}\left\|\mathbf{R}_k - \widehat{\mathbf{X}}_k\right\|_F^2 + \bar{t}_X. \tag{49}$$

The next scaling factor $\kappa^{(i+1)}$ is obtained by averaging $\kappa^{(i)}$ and $v_{\text{tar}}^{(i)}/\bar{v}^{(i)}$ with weights $1-\eta$ and η, respectively.

The input to MFLE is damped in a similar manner. In the feedback updates (28)–(29), $b = \bar{t}_X/v$ is clipped to $[\varepsilon, 1-$

$\varepsilon]$, where $0 < \varepsilon < 1/2$. With the same damping factor, the feedback messages for $i \geq 2$ are

$$\mathbf{X}_k^{\mathrm{pri},(i)} = (1-\eta)\mathbf{X}_k^{\mathrm{pri},(i-1)} + \eta \mathbf{X}_{k,\mathrm{out}}^{(i)}, \tag{50}$$

$$\gamma^{(i)} = (1-\eta)\gamma^{(i-1)} + \eta \gamma_{\mathrm{out}}^{(i)}. \tag{51}$$

C. Computational Complexity Analysis

With precomputed chirp factors, L AFDM frames require at most $LC_{\mathrm{FFT}}(N) + 2LN$ complex multiplications [11], where $C_{\mathrm{FFT}}(n)$ is the cost of an n-point fast Fourier transform (FFT) or inverse FFT. For the same LN useful symbols, rectangular-pulse OTFS without transmit windowing requires $NC_{\mathrm{FFT}}(L)$ after cancelling the consecutive discrete Fourier transform (DFT) and inverse DFT operations [36].

The computational complexity of the receiver is primarily determined by the operations in MFLE and GPJD, together with parameter estimation. With fixed phases and $Q \leq KN$, MFLE can be implemented using the singular value decomposition (SVD), $\widehat{\mathbf{A}} = \mathbf{U}\,\mathrm{diag}(s_1, \ldots, s_Q)\mathbf{V}^{\mathsf{H}}$, which is reused across iterations. Here s_a are the singular values, and the scalar updates in (13)–(14) can be written as

$$f_{\mathrm{L}} = \frac{1}{KN}\sum_{a=1}^{Q} \frac{\gamma s_a^2}{\widehat{\sigma}^2 + \gamma s_a^2}, \tag{52}$$

$$t_{\mathrm{L}} = \gamma(1 - f_{\mathrm{L}}), \tag{53}$$

$$v_{\mathrm{L}} = \frac{t_{\mathrm{L}}}{f_{\mathrm{L}}}. \tag{54}$$

Here $\mathbf{R}_{\mathrm{L}} = \mathbf{X}^{\mathrm{pri}} + (\mathbf{X}_{\mathrm{L}} - \mathbf{X}^{\mathrm{pri}})/f_{\mathrm{L}}$.

For each block, the SVD and noise-variance estimation require $O(KNQ^2)$ and $O(QNM^2 + M^3)$ operations, respectively. The phase-estimation cost is denoted by C_{phase}, with $C_{\mathrm{phase}} = 0$ for known phases.

For each iteration, the MFLE transformations involve $O(QKMN^2)$ operations, while the spatial covariance operations and matrix factorizations require $O(KM^2N^2)$ and $O(M^3)$ operations, respectively. The per-device costs of a forward pass through the denoiser and classifier are denoted by C_θ and C_ϕ, respectively. Each GPJD update evaluates the denoiser $1+2L_z$ times and the classifier once per device. Consequently, these forward passes require $K[(1+2L_z)C_\theta + C_\phi]$ operations.

Overall, the online complexity for I iterations is $O(C_{\mathrm{phase}} + QNM^2 + M^3 + KNQ^2 + IC_{\mathrm{iter}})$, where C_{iter} denotes the total per-iteration cost. The pilot matrix, SVD factors, channel messages, and spatial covariances require $O(QKN + KMN^2 + KM^2)$ memory.

VI. Simulation Results

A. Simulation Settings

1) Channel Dataset and System Parameters: The default system parameters are listed in Table I. Devices are uniformly distributed within the satellite beam, and their delays, Doppler shifts, and array responses follow the satellite-device geometry. Large-scale effects are compensated. Channels follow the NTN-TDL-D profile [3], with delay spread uniformly distributed over $[30, 50]$ ns. The line-of-sight gain has fixed amplitude and uniform phase, and the diffuse gains are proper complex Gaussian. We generate $50{,}000$ training channels and $5{,}000$ validation channels of size $16 \times 16 \times 16$.

TABLE I
Simulation Parameters

Parameter	Value
Carrier frequency f_c	2 GHz
Satellite altitude	600 km
Beam radius	128 km
Minimum elevation angle	30°
UPA dimensions $M_y \times M_z$	4×4
Array element spacing d_y, d_z	$\lambda/2$
Number of chirp subcarriers N	16
Subcarrier spacing Δf	30 kHz
Chirp parameters (c_1, c_2)	$(7/32, 1/320)$
CPP length N_{CPP}	336
Differential-delay range	$[0, 699]$ μs
Doppler-shift range	$[-41, 41]$ kHz
Power delay profile	NTN-TDL-D
Delay spread	$[30, 50]$ ns
Rician factor	11.707 dB
Default (K, Q, p)	$(40, 16, 0.1)$

For $Q \leq KN$, the pilots are constructed from distinct DFT rows $r_q \in \{0, \ldots, KN-1\}$, with a column permutation $\{c_{kn}\}_{k,n}$ of the same set and independent phases $\phi_{kn} \sim \mathcal{U}[0, 2\pi)$:

$$[\mathbf{p}_{qk}]_n = \sqrt{\frac{E_p}{N}} e^{-\mathrm{j}2\pi r_q c_{kn}/(KN)} e^{\mathrm{j}\phi_{kn}}, \tag{55}$$

with $\sum_{n=0}^{N-1} |[\mathbf{p}_{qk}]_n|^2 = E_p$.

The received SNR is defined as the average signal power over the QMN entries of $\mathbf{AX}$ divided by σ^2. We set σ^2 separately for each scene to obtain the specified SNR.

2) Network Architectures and Training: The denoiser uses the RefineNet [37] variant in [13], with $2M$ real and imaginary input maps over $N \times N$ DAFT coordinates. Each stage contains two residual blocks, each with two 3×3 convolutions, conditional instance normalization, and exponential linear unit (ELU) activations. The stage outputs are combined by four refinement blocks that restore the spatial resolution, with 3×3 convolutions at the network input and output. Table II lists the channel widths and spatial resolutions of both networks.

Noise conditioning uses an embedding of $\log(v)/5$, formed by two 64-unit fully connected layers with sigmoid-weighted linear unit (SiLU) activations and linearly projected to the normalization layers. The denoiser also adopts the preconditioning in [38]:

$$\begin{aligned} \mathbf{f}_\theta(\mathbf{u}, v) &= c_{\mathrm{skip}}(v)\mathbf{u} + c_{\mathrm{out}}(v)\mathbf{g}_\theta(c_{\mathrm{in}}(v)\mathbf{u}, v), \\ c_{\mathrm{in}}(v) &= (\sigma_{\mathrm{d}}^2 + v/2)^{-1/2}, \\ c_{\mathrm{skip}}(v) &= \frac{\sigma_{\mathrm{d}}^2}{\sigma_{\mathrm{d}}^2 + v/2}, \\ c_{\mathrm{out}}(v) &= \left(\frac{\sigma_{\mathrm{d}}^2 v/2}{\sigma_{\mathrm{d}}^2 + v/2}\right)^{1/2}. \end{aligned} \tag{56}$$

Here $\sigma_{\mathrm{d}}^2 = 1/2$ is the nominal variance per real coordinate after normalization, and $\mathbf{g}_\theta$ is the RefineNet described above.

TABLE II
NETWORK ARCHITECTURES

Layer or stage	Channels	Map size
Channel denoiser		
Input convolution	128	$N \times N$
Residual stage 1	128	$N \times N$
Residual stage 2	256	$N/2 \times N/2$
Residual stage 3 (dilation 2)	256	$N/2 \times N/2$
Residual stage 4 (dilation 4)	256	$N/2 \times N/2$
Refinement blocks 1–2	$256, 256$	$N/2 \times N/2$
Refinement block 3	128	$N/2 \times N/2$
Refinement block 4	128	$N \times N$
Output convolution	$2M$	$N \times N$
Activity classifier		
Convolutional stage 1	64	$N \times N$
Convolutional stage 2	128	$N/2 \times N/2$
Convolutional stage 3	128	$N/4 \times N/4$
Fully connected layers	$128, 1$	—

The classifier estimates the log likelihood ratio from $4M$ real and imaginary maps of $\mathbf{R}$ and $\mathbf{F}_\theta(\mathbf{R}, v)$, scaled by $\sqrt{2/(1+v)}$, together with five scalars: $\log(v)/5$, logarithmic transforms of input/output/residual energies normalized by Dv, and a sign-preserving logarithmic transform of the real inner product normalized by Dv. Its three convolutional stages each contain two 3×3 convolutions with SiLU activations and padding 1. Downsampling uses stride 2 in the first convolution of stages 2–3; the remaining convolutions use stride 1. Noise conditioning is provided by a 64-unit fully connected SiLU embedding of $\log(v)/5$, whose linear projections are added to the stage outputs. The flattened features and five scalars are combined by the fully connected layers in Table II, with SiLU between them, to produce one logit per device.

We train the denoiser by minimizing (33) with the preconditioning in (56). The variance v is sampled uniformly on a logarithmic scale over $[10^{-3}, 10^{2}]$. We use Adam for 200 epochs with batch size 64, initial learning rate 10^{-4}, and exponential moving average decay 0.999. STMP uses the same clean channels and number of epochs.

With the denoiser fixed, we train the classifier using (39) for 100 epochs, with batch size 32 and initial learning rate 3×10^{-4}. For both networks, we select the model with the lowest validation loss.

3) Compared Receivers: We compare GP-MP with five baselines. GP-MP-EP uses estimated phases; the other configurations use known phases:

- **STMP:** We adapt STMP [1] to the multi-frame AFDM model, retaining its white-noise-trained first- and second-order score networks for conditional mean and variance estimation. We replace its Gaussian-message-based activity-probability update with a learned density-ratio estimator trained using the corresponding first-order network.
- **Simultaneous orthogonal matching pursuit (SOMP)** [39]: This method selects dictionary atoms shared by the receive antennas and estimates their coefficients by least squares. We use an integer delay–Doppler dictionary and select up to 64 atoms.
- **GAMP-SBL:** This method uses generalized approximate message passing in sparse Bayesian learning [40]. Our implementation uses the integer delay–Doppler dictionary, shares row precisions across receive antennas, and allows up to 1,000 iterations.
- **GNN-ASPL:** This method alternates AMP-based channel estimation with adaptive sparsity pattern learning using a graph neural network [10]. It uses the same integer delay–Doppler dictionary and a 10-layer network.
- **Off-grid SBL:** In addition to the channel coefficients, this method estimates continuous delay and Doppler offsets through a first-order linear approximation of the dictionary [41]. We allow up to 120 iterations. For the SNR comparison, the 7, 13, and 25 Doppler grid points are paired with 12, 24, and 48 retained candidates per device and 48, 96, and 192 offset candidates per iteration, respectively; all three configurations use 112 delay grid points.

4) Evaluation Settings and Metrics: Phase estimation in GP-MP-EP covers $d \in [0, N)$ and $|\nu| \le \nu_{\max}$, where $\nu_{\max}$ is the channel model's Doppler bound. Pilot correlations over 128 phase bins identify three coarse candidates. The Doppler values corresponding to each candidate phase within this bound form the initial search set, from which the best 12 are refined by at most 25 L-BFGS-B iterations per candidate.

All receivers estimate the noise variance from the observations. STMP and GP-MP use an activity threshold of 0.5 and default damping factors of 0.2 and 0.4, respectively. Their iterations are bounded by $I_{\min} = 60$ and $I_{\max} = 600$, with a relative-change tolerance of 5×10^{-4}. After $I_{\min}$ iterations, convergence is assessed from the stability of channel and activity estimates and the consistency of variance updates.

GP-MP uses $L_z = 1$ and $\delta_{\mathrm{FD}} = 10^{-2}$ in (37). The average conditional variance has a lower bound $\tau_{\min} = 10^{-4}$.

The evaluation protocol specifies 1,000 Monte Carlo trials per operating point.

For scenes $e \in \mathcal{T}$, the normalized mean-squared error (NMSE) is the total squared Frobenius error between $\widehat{\mathbf{X}}_{ek}$ and $\mathbf{X}_{ek}$ divided by the total squared Frobenius norm of $\mathbf{X}_{ek}$. Both totals include all scenes in $\mathcal{T}$ and all K devices; the ratio is expressed in decibels. For channel denoising, the same metric uses $\mathbf{H}$ in place of $\mathbf{X}$ and is accumulated over individual channel samples. The activity error rate (AER) is the total number of missed detections and false alarms divided by $K|\mathcal{T}|$.

B. Channel Denoising Performance

We first evaluate the channel denoisers under $\mathbf{R} = \mathbf{H} + \mathbf{W}$, where $\mathrm{vec}(\mathbf{W}) \sim \mathcal{CN}(\mathbf{0}, v\mathbf{I})$ and v is known. The nominal SNR is $-10 \log_{10} v$ and takes six values from -20 to 30 dB. The reference estimators are:

- **Least squares (LS):** The estimate is the noisy observation $\mathbf{R}$.
- **Gaussian-prior estimate:** The estimate is $\widehat{P}_H \mathbf{R}/(\widehat{P}_H + v)$, where $\widehat{P}_H$ is the training-channel power.
- **Genie-aided conditional MMSE:** We compute the analytic conditional MSE of the remaining Gaussian gains given the true path geometry, powers, and line-of-sight gain.

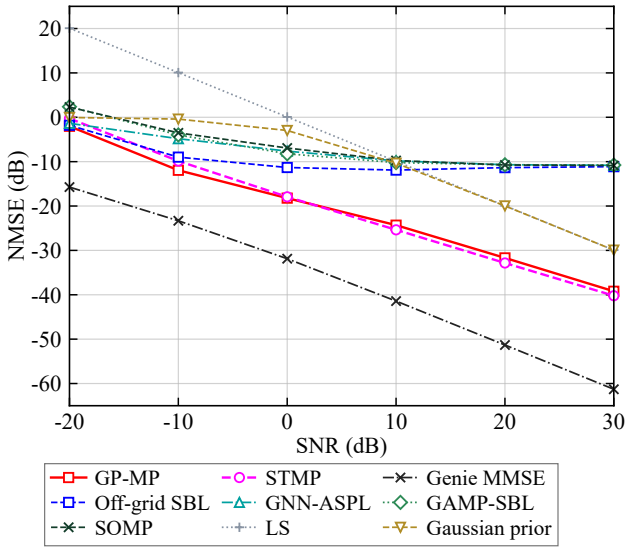


Fig. 4. Channel denoising under AWGN with known variance.

TABLE III
ACTIVITY DETECTION PERFORMANCE AND ITERATION-LIMIT RATES

	STMP				GP-MP			
	Gaussian		Density ratio		Gaussian		Density ratio	
SNR (dB)	AER (%)	Limit (%)	AER (%)	Limit (%)	AER (%)	Limit (%)	AER (%)	Limit (%)
−4	9.225	2	0.525	0	44.575	63	0.200	0
−2	7.875	3	0.225	0	43.475	55	0.125	0
0	6.375	2	0.100	0	40.975	46	0.075	0
2	5.550	1	0.050	0	36.750	31	0.025	0
4	4.800	1	0.025	0	32.050	19	0.025	0
6	4.225	0	0.025	0	27.425	11	0.025	0

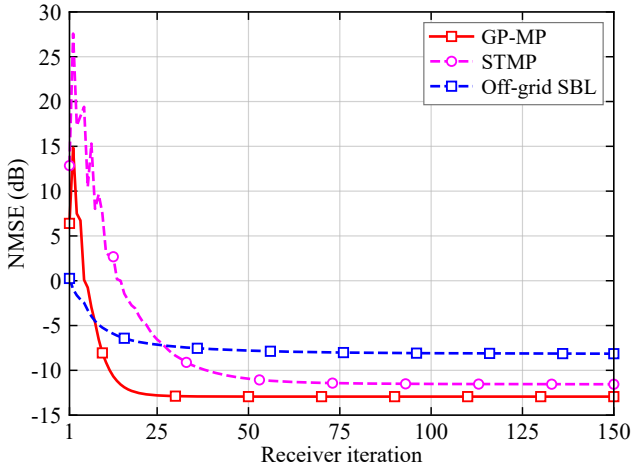


Fig. 5. NMSE versus iteration at 0 dB, with $K = 40$, $Q = 16$, and $p = 0.1$.

We also evaluate the four sparse-recovery baselines in this denoising setting.

Figure 4 shows that both learned denoisers achieve lower NMSE than LS and the Gaussian-prior estimate throughout the tested SNR range. Their advantage over the sparse-recovery baselines increases at moderate and high SNR, where these baselines approach error floors while the NMSE of the learned denoisers continues to decrease. The GP-MP denoiser yields lower NMSE at low SNR, whereas the STMP denoiser yields lower NMSE at high SNR. Both remain above the genie-aided reference.

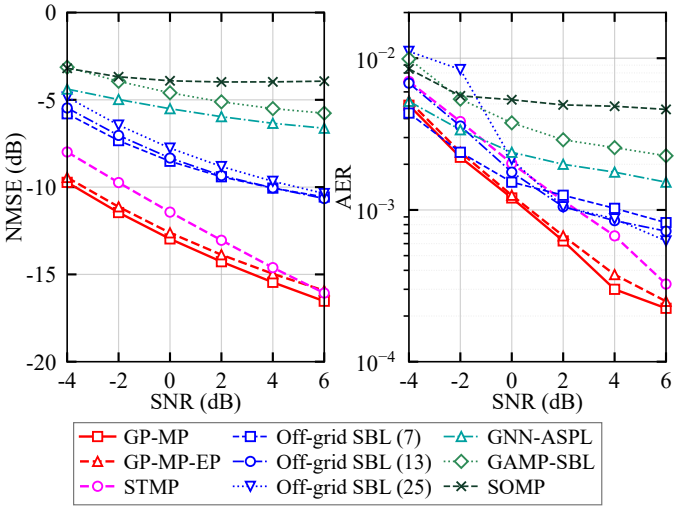


Fig. 6. Channel NMSE and AER versus SNR with $K = 40$, $Q = 16$, and $p = 0.1$.

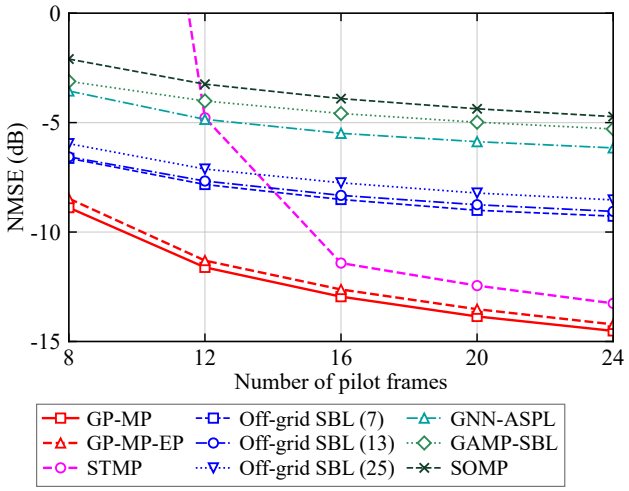


Fig. 7. Channel NMSE versus the number of pilot frames at 0 dB, with $K = 40$ and $p = 0.1$.

C. Effect of Density Ratio Estimation

To assess density ratio estimation, we compare the learned activity update with the Gaussian approximation

$$\ell_k^{\mathrm{G}} = \frac{\|\mathbf{R}_k\|_F^2}{v} - \frac{\|\mathbf{R}_k - \mathbf{E}_k\|_F^2}{\tau_k^{\mathrm{ext}} + v} - D \log\left(1 + \frac{\tau_k^{\mathrm{ext}}}{v}\right), \tag{57}$$

$$p_k = \mathrm{sigmoid}(\mathrm{logit}\, \rho + \ell_k^{\mathrm{G}}). \tag{58}$$

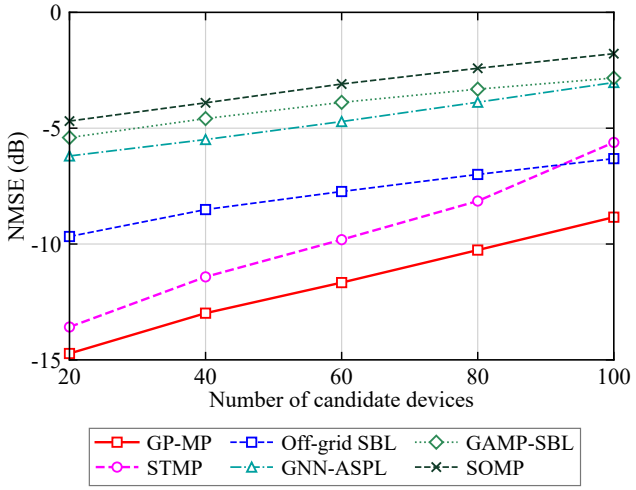


Fig. 8. Channel NMSE versus the number of candidate devices at 0 dB, with $Q = 16$ and $p = 0.1$.

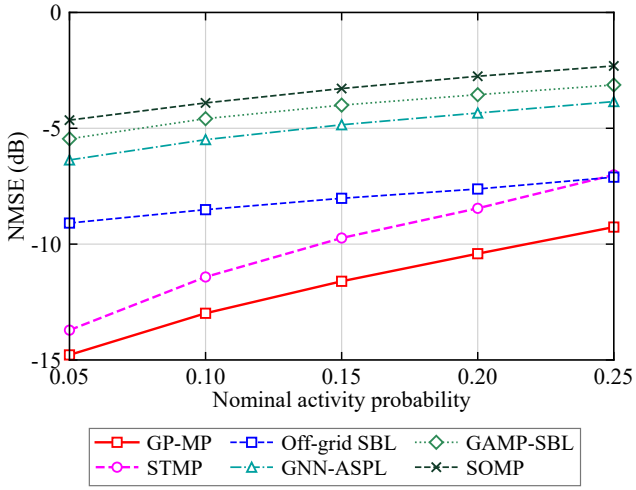


Fig. 9. Channel NMSE versus nominal activity probability at 0 dB, with $K = 40$ and $Q = 16$.

For both STMP and GP-MP, we change only the activity update and keep the remaining settings fixed. Table III reports the AER over all runs and the percentage reaching the 600-iteration limit (Limit).

Table III shows that, for both receivers, the learned activity update yields lower AER than the Gaussian update at every tested SNR. No run using the learned update reaches the iteration limit, whereas some runs using the Gaussian update do. The improvement is larger for GP-MP, although STMP performs better under the Gaussian update. These results support the effectiveness of density ratio estimation for activity detection under the learned channel prior.

D. JADCE Performance

Unless stated otherwise, the default parameters in Table I are used.

Figure 5 compares the NMSE trajectories with early stopping disabled. The GP-MP curve stabilizes earlier and at a lower NMSE than STMP. Off-grid SBL also levels off, but with a larger estimation error. GP-MP therefore achieves a more accurate channel estimate within fewer iterations than STMP.

Figure 6 compares channel NMSE and AER over SNRs from -4 to 6 dB. GP-MP achieves the lowest NMSE throughout this range, with an approximately 1.5 dB advantage over STMP at 0 dB. Its advantage over STMP decreases as SNR increases, whereas the gap to the sparse-recovery baselines widens. Off-grid SBL is the most accurate of the four sparse-recovery baselines, and GP-MP outperforms all three Off-grid SBL configurations.

The AER also generally decreases with SNR. GP-MP has lower AER than STMP throughout this SNR range and outperforms the four sparse-recovery baselines at moderate and high SNR. These results demonstrate improved JADCE performance with a denoiser that provides channel estimates and average variance without a separately trained variance model.

With estimated phases, GP-MP-EP incurs an NMSE increase of 0.28–0.60 dB relative to GP-MP, while its AER remains close to the known-phase result. This comparison shows that estimating the phases from pilot observations retains most of the known-phase receiver's performance over this SNR range.

Figure 7 shows channel NMSE as Q increases from 8 to 24. With fixed energy per frame, additional frames provide more observations and generally reduce NMSE. GP-MP and GP-MP-EP achieve lower NMSE than all baselines at each frame count. For an NMSE target of -12 dB, both reach the target at 16 frames, compared with 20 frames for STMP, corresponding to a 20% reduction in pilot frames.

STMP exhibits severe degradation at low pilot counts: its NMSE is 34.70 dB at $Q = 8$ and -4.78 dB at $Q = 12$, compared with -8.88 dB and -11.61 dB for GP-MP, respectively. This pronounced loss indicates that STMP is more sensitive to limited pilot observations. GP-MP-EP incurs an NMSE increase of 0.30–0.41 dB relative to GP-MP across these frame counts.

Figure 8 compares NMSE as the number of candidate devices increases from $K = 20$ to 100. With the activity probability and pilot-frame count fixed, both the number of candidate channels and the expected number of active devices increase. The NMSE therefore increases for all methods. GP-MP maintains the lowest NMSE, with a growing advantage over STMP as K increases. This demonstrates its effectiveness at higher device loads with the available pilot and antenna resources.

Figure 9 shows NMSE for nominal activity probabilities from 0.05 to 0.25. Increasing the activity probability increases the expected number of active devices, while the numbers of candidate devices, pilot frames, and receive antennas remain fixed. The NMSE increases for all methods. GP-MP maintains the lowest NMSE, with a smaller increase than STMP. Its larger advantage at higher activity probabilities indicates better tolerance to increasing activity loads.

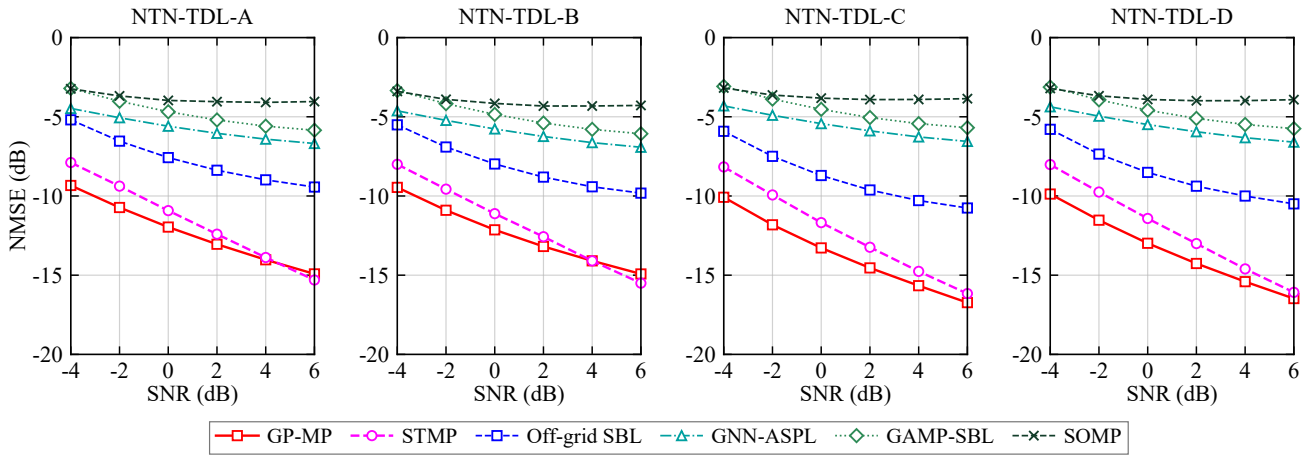


Fig. 10. Channel NMSE for NTN-TDL profiles A, B, C, and D (from left to right), with $K = 40$, $Q = 16$, and $p = 0.1$.

E. Comparison Across Channel Profiles

To examine performance under channel mismatch, we apply the networks trained on NTN-TDL-D to profiles A, B, and C without retraining.

Figure 10 shows that GP-MP outperforms the four sparse-recovery baselines on all four profiles. This indicates that the learned prior remains effective across these channel profiles. The relative performance of the two learned receivers depends on the profile and SNR; STMP is more accurate on profiles A and B at 6 dB.

VII. Conclusion

In this paper, we investigated GNSS-independent GFRA for LEO satellite IoT systems. We proposed a multi-frame AFDM transmission scheme that accommodates large differential delays and Doppler shifts without device-side precompensation and derived its input-output relationship. Additional pilot frames provide more observations without increasing the number of channel coefficients to be estimated. We learned a score-based generative prior for AFDM effective channels without imposing sparse or grid-aligned channel assumptions, and developed a denoiser that provides channel estimates and average variance without requiring a separately trained variance model. We incorporated this denoiser into a message-passing JADCE receiver comprising MFLE and GPJD. The learned density-ratio estimator in GPJD enables activity detection under the learned prior without explicit marginal-likelihood evaluation. Simulation results demonstrated improved activity detection and channel estimation over the adapted STMP baseline, with fewer pilot frames required for comparable channel-estimation performance. The receiver also achieved lower NMSE than the sparse-recovery baselines across the tested NTN-TDL profiles without retraining.

References

[1] C. Cai, W. Jiang, X. Yuan, and Y.-J. A. Zhang, "Joint activity detection and channel estimation for massive connectivity: Where message passing meets score-based generative priors," *IEEE J. Sel. Areas Commun.*, vol. 44, pp. 3575–3590, 2026.

[2] Y. Xu, L. Xiao, C. Ding, P. Xiao, M. Wen, and T. Jiang, "AFDM-aided grant-free random access for LEO SIoT: Performance analysis and near-optimal joint detection," *IEEE Trans. Commun.*, vol. 74, pp. 840–853, 2026.

[3] B. Shen, Y. Wu, W. Zhang, S. Chatzinotas, and B. Ottersten, "LEO satellite-enabled random access with large differential delay and doppler shift," *IEEE Trans. Wireless Commun.*, vol. 24, no. 4, pp. 2876–2893, 2025.

[4] O. Kodheli *et al.*, "Satellite communications in the new space era: A survey and future challenges," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 70–109, 2021.

[5] Y. Mei *et al.*, "Compressive sensing-based joint activity and data detection for grant-free massive IoT access," *IEEE Trans. Wireless Commun.*, vol. 21, no. 3, pp. 1851–1869, 2022.

[6] 3GPP, "NR; NR and NG-RAN overall description; stage-2," 3rd Generation Partnership Project, Technical Specification 38.300, Oct. 2022, version 17.2.0, Release 17. [Online]. Available: https://www.etsi.org/deliver/etsi_ts/138300_138399/138300/17.02.00_60/ts_138300v170200p.pdf

[7] ——, "Study on satellite access - phase 3," 3rd Generation Partnership Project, Technical Report 22.865, Dec. 2023, version 19.2.0, Release 19. [Online]. Available: https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=4089

[8] Y. Huang, Y. Guo, C. Guo, Y. Wang, and X. Guo, "Prior-agnostic AFDM channel estimation for high-mobility LEO satellite communications using diffusion models," *IEEE Wireless Commun. Lett.*, vol. 15, pp. 2949–2953, 2026.

[9] R. Hadani *et al.*, "Orthogonal time frequency space modulation," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, 2017, pp. 1–6.

[10] C. Ding, L. Xiao, J. Zhou, L. Zhu, P. Xiao, and T. Jiang, "Learning sparsity pattern: An adaptive joint detection for AFDM grant-free system in LEO satellites communication," *IEEE Trans. Veh. Technol.*, 2026, early access, doi: 10.1109/TVT.2026.3729877.

[11] A. Bemani, N. Ksairi, and M. Kountouris, "Affine frequency division multiplexing for next generation wireless communications," *IEEE Trans. Wireless Commun.*, vol. 22, no. 11, pp. 8214–8229, 2023.

[12] M. Ke, Z. Gao, Y. Wu, X. Gao, and R. Schober, "Compressive sensing-based adaptive active user detection and channel estimation: Massive access meets massive MIMO," *IEEE Trans. Signal Process.*, vol. 68, pp. 764–779, 2020.

[13] Y. Song and S. Ermon, "Generative modeling by estimating gradients of the data distribution," in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.

[14] M. Arvinte and J. I. Tamir, "MIMO channel estimation using score-based generative models," *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 3698–3713, 2023.

[15] K. Ying, Z. Gao, S. Chen, T. Q. S. Quek, and H. V. Poor, "Generative diffusion model driven massive random access in massive MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 8210–8227, 2026.

[16] A. Menon and C. S. Ong, "Linking losses for density ratio and class-probability estimation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 48, 2016, pp. 304–313.

[17] F. D. Neeser and J. L. Massey, "Proper complex random processes with applications to information theory," *IEEE Trans. Inf. Theory*, vol. 39, no. 4, pp. 1293–1302, Jul. 1993.

[18] K.-X. Li, X. Gao, and X.-G. Xia, "Channel estimation for LEO satellite massive MIMO OFDM communications," *IEEE Trans. Wireless Commun.*, vol. 22, no. 11, pp. 7537–7550, 2023.

[19] A. Bemani, "Affine frequency division multiplexing (AFDM) for wireless communications," Ph.D. dissertation, Sorbonne Université, 2023. [Online]. Available: https://www.eurecom.eu/publication/7447/download/comsys-publi-7447.pdf

[20] T. P. Minka, "Expectation propagation for approximate Bayesian inference," in *Proc. Conf. Uncertainty Artif. Intell. (UAI)*, 2001, pp. 362–369.

[21] J. Ma, X. Yuan, and L. Ping, "Turbo compressed sensing with partial DFT sensing matrix," *IEEE Signal Process. Lett.*, vol. 22, no. 2, pp. 158–161, Feb. 2015.

[22] J. Ma and L. Ping, "Orthogonal AMP," *IEEE Access*, vol. 5, pp. 2020–2033, 2017.

[23] A. Vehtari *et al.*, "Expectation propagation as a way of life: A framework for Bayesian inference on partitioned data," *J. Mach. Learn. Res.*, vol. 21, no. 17, pp. 1–53, 2020.

[24] L. Liu and W. Yu, "Massive connectivity with massive MIMO—Part I: Device activity detection and channel estimation," *IEEE Trans. Signal Process.*, vol. 66, no. 11, pp. 2933–2946, 2018.

[25] B. Efron, "Tweedie's formula and selection bias," *J. Amer. Stat. Assoc.*, vol. 106, no. 496, pp. 1602–1614, 2011.

[26] A. Dytso, H. V. Poor, and S. Shamai (Shitz), "Conditional mean estimation in Gaussian noise: A meta derivative identity with applications," *IEEE Trans. Inf. Theory*, vol. 69, no. 3, pp. 1883–1898, 2023.

[27] P. Vincent, "A connection between score matching and denoising autoencoders," *Neural Comput.*, vol. 23, no. 7, pp. 1661–1674, 2011.

[28] H. Manor and T. Michaeli, "On the posterior distribution in denoising: Application to uncertainty quantification," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[29] C. A. Metzler, A. Maleki, and R. G. Baraniuk, "From denoising to compressed sensing," *IEEE Trans. Inf. Theory*, vol. 62, no. 9, pp. 5117–5144, 2016.

[30] M. Wax and T. Kailath, "Detection of signals by information theoretic criteria," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 33, no. 2, pp. 387–392, 1985.

[31] R. R. Nadakuditi and A. Edelman, "Sample eigenvalue based detection of high-dimensional signals in white noise using relatively few samples," *IEEE Trans. Signal Process.*, vol. 56, no. 7, pp. 2625–2638, 2008.

[32] M. E. Tipping and A. C. Faul, "Fast marginal likelihood maximisation for sparse bayesian models," in *Proc. 9th Int. Workshop Artif. Intell. Statist. (AISTATS)*, 2003, pp. 276–283. [Online]. Available: https://proceedings.mlr.press/r4/tipping03a.html

[33] C. Zhu, R. H. Byrd, P. Lu, and J. Nocedal, "Algorithm 778: L-BFGS-B: Fortran subroutines for large-scale bound-constrained optimization," *ACM Trans. Math. Softw.*, vol. 23, no. 4, pp. 550–560, 1997.

[34] S. Rangan, P. Schniter, and A. K. Fletcher, "Vector approximate message passing," *IEEE Trans. Inf. Theory*, vol. 65, no. 10, pp. 6664–6684, 2019.

[35] J. P. Vila and P. Schniter, "Expectation-maximization Gaussian-mixture approximate message passing," *IEEE Trans. Signal Process.*, vol. 61, no. 19, pp. 4658–4672, Oct. 2013.

[36] T. Zou, W. Xu, H. Gao, Z. Bie, Z. Feng, and Z. Ding, "Low-complexity linear equalization for OTFS systems with rectangular waveforms," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, 2021, pp. 1–6.

[37] G. Lin, A. Milan, C. Shen, and I. Reid, "RefineNet: Multi-path refinement networks for high-resolution semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 5168–5177.

[38] T. Karras, M. Aittala, T. Aila, and S. Laine, "Elucidating the design space of diffusion-based generative models," in *Adv. Neural Inf. Process. Syst.*, vol. 35, 2022.

[39] J. A. Tropp, A. C. Gilbert, and M. J. Strauss, "Algorithms for simultaneous sparse approximation. part I: Greedy pursuit," *Signal Process.*, vol. 86, no. 3, pp. 572–588, 2006.

[40] X. Zhang, P. Fan, L. Hao, and X. Quan, "Generalized approximate message passing based bayesian learning detectors for uplink grant-free NOMA," *IEEE Trans. Veh. Technol.*, vol. 72, no. 11, pp. 15 057–15 061, 2023.

[41] Z. Wei, W. Yuan, S. Li, J. Yuan, and D. W. K. Ng, "Off-grid channel estimation with sparse bayesian learning for OTFS systems," *IEEE Trans. Wireless Commun.*, vol. 21, no. 9, pp. 7407–7426, 2022.